Beyond the Dashboard:
A Conceptual Framework for Evaluating Insight Quality in AI-Augmented Business Intelligence Systems

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ABSTRACT

The integration of artificial intelligence into business intelligence (BI) platforms has given rise to augmented analytics: the automated detection, generation, and narration of data-driven insights with minimal human analytical intervention. Platforms increasingly surface algorithmically generated observations—anomalies, correlations, forecasts, and natural-language summaries—directly within managerial dashboards, reducing reliance on dedicated data analysts. While this promises faster, more scalable decision support, it introduces a largely unaddressed problem: organizations lack systematic criteria for evaluating whether these automatically generated insights are statistically sound, contextually relevant, non-trivial, and actionable, as opposed to spurious, redundant, or misleading. Existing BI literature has extensively studied dashboard usability, visualization design, and user adoption, while data quality research has focused predominantly on input data rather than generated output. Insight quality itself—as a distinct analytical construct—remains conceptually underdeveloped. This paper addresses this gap by synthesizing literature from augmented analytics, automated insight discovery, data mining evaluation metrics, and decision support systems to propose an integrative conceptual framework—the Insight Quality Evaluation (IQE) framework—comprising four evaluative dimensions: statistical validity, contextual relevance, novelty, and actionability. The framework further incorporates task complexity and organizational data maturity as moderating factors shaping insight quality outcomes. The paper proposes a mixed-methods research design, combining computational evaluation of insight-generation algorithms with expert-panel validation, for future empirical testing. This work contributes a structured vocabulary and evaluative lens for augmented analytics research and offers BI practitioners actionable criteria for auditing AI-generated insights before they inform organizational decisions.

Keywords: Augmented Analytics; Automated Insight Generation; Business Intelligence; Insight Quality; Data-Driven Decision-Making; Artificial Intelligence in Analytics; Decision Support Systems; Data Mining Evaluation

1. INTRODUCTION

Organizations have entered a third distinct wave of business intelligence (BI) evolution. The first wave, spanning the 1990s and 2000s, centered on IT-managed reporting and static, centrally produced dashboards. The second wave, emerging in the 2010s, introduced self-service BI, empowering business users to explore data independently through drag-and-drop visualization tools such as Tableau and Power BI. The third and current wave—commonly termed augmented analytics—embeds artificial intelligence (AI) and machine learning (ML) directly into the analytics pipeline, automating not only data preparation but also insight generation and insight explanation. In this paradigm, dashboards no longer merely display data that a human analyst has already interpreted; instead, algorithms actively surface correlations, anomalies, forecasts, and natural-language narratives with little or no human analytical intervention. This shift has moved rapidly from research curiosity to mainstream commercial practice. Contemporary BI platforms—including Microsoft Power BI, Tableau, Qlik, Looker, ThoughtSpot, and Domo—now market automated insight generation as a core differentiator rather than an experimental add-on.

Gartner itself has characterized this machine-driven, low-human-intervention mode of BI as the current and likely durable state of the industry.

The promise of augmented analytics is considerable: faster time-to-insight, reduced dependency on scarce data science talent, and broader organizational access to analytical reasoning. Yet this promise rests on an underexamined assumption—that the insights automatically surfaced by these systems are, in fact, good insights. An "insight" generated by a pattern-detection algorithm may be statistically valid but practically trivial; it may be numerically significant but contextually meaningless; it may be superficially novel but merely a restatement of a well-known seasonal effect; or it may point to a correlation with no clear implication for action. The data mining literature has long recognized that automated pattern discovery over large datasets tends to produce an overwhelming number of candidate patterns, only a small fraction of which are genuinely meaningful to a human decision-maker (Geng & Hamilton, 2006). This problem, well studied within data mining as the challenge of interestingness measurement, has not been adequately translated into the applied context of organizational BI dashboards, where automatically generated insights are increasingly consumed directly by managers with limited capacity—or limited inclination—to independently audit the underlying analytical logic.

This creates a consequential organizational risk. Where earlier generations of BI required a human analyst to interpret data and construct a narrative—implicitly filtering out spurious or low-value observations through domain expertise—augmented analytics systems increasingly bypass this human filtering step. If the insight-generation layer itself does not incorporate a robust standard for what constitutes a valid, relevant, novel, and actionable insight, organizations risk institutionalizing decision-making on the basis of statistical noise, decontextualized correlations, or trivial restatements of already-known facts. As organizations scale the deployment of augmented analytics across operational, financial, and strategic dashboards, an accumulation of low-quality automated insights, if uncritically trusted, could measurably degrade rather than enhance decision quality—precisely the opposite of the technology's stated purpose.

Despite the scale of this practical exposure, the scholarly literature has not developed an integrated framework for evaluating the quality of insights generated by augmented analytics systems as a distinct analytical construct. Three adjacent bodies of literature have developed largely in isolation from one another. First, the data mining and knowledge discovery literature has produced a mature body of work on interestingness measures—metrics such as support, confidence, lift, and novelty used to rank discovered patterns (Geng & Hamilton, 2006)—and more recent computational frameworks for automated insight mining in multidimensional data, such as Microsoft's QuickInsights system (Ding et al., 2019). Second, the BI and information systems literature has extensively examined dashboard design, usability, and adoption, and has established links between BI tool implementation and decision-making effectiveness. Third, an emerging and still sparse body of work has begun to examine user trust and cognition in relation to augmented analytics, highlighting that scholarly attention to how business users engage with automatically generated analytical content remains comparatively limited despite growing practitioner adoption. The result is a fragmented literature in which no single framework connects the computational question of how insights are algorithmically identified, the organizational question of how they are consumed through dashboards, and the evaluative question of what makes an automatically generated insight good enough to inform managerial decisions. This paper addresses that gap directly.

This study makes three contributions. First, it synthesizes and critically integrates literature from data mining interestingness research, BI dashboard effectiveness research, and augmented analytics/decision support research to expose the conceptual gap described above. Second, it proposes an integrative conceptual framework—the Insight Quality Evaluation (IQE) framework—that defines insight quality in AI-augmented BI systems along four dimensions: statistical validity, contextual relevance, novelty, and actionability, moderated by task complexity and organizational data maturity. Third, recognizing that this is a conceptual contribution rather than an empirical study, the paper proposes a rigorous mixed-methods research design that future researchers can operationalize to empirically validate and refine the framework using publicly available datasets.

The remainder of the paper proceeds as follows. Section 2 critically reviews the relevant literature across the three converging domains. Section 3 articulates the specific research gap. Sections 4 and 5 present the research questions and objectives. Section 6 develops the theoretical background underpinning the framework. Section 7 introduces the conceptual framework itself. Sections 8 and 9 detail the proposed methodology and data analysis approach for future empirical validation. Sections 10 through 13 discuss implications, contributions, limitations, and future research directions, followed by the conclusion.

2. LITERATURE REVIEW

The literature relevant to this study spans four historically distinct but increasingly convergent domains: (a) decision support systems (DSS) and business intelligence theory; (b) automated insight discovery and interestingness measurement in data mining; (c) visual analytics research on insight provenance and quality; and (d) natural language generation (NLG) as applied to data narration. This section critically reviews each domain, highlighting where findings converge, where they conflict, and where the connective theoretical tissue is missing.

2.1 Theoretical Roots: From Decision Support Systems to Augmented Analytics

The conceptual lineage of BI systems traces to Sprague's (1980) foundational framework for decision support systems, which characterized DSS around a dialogue–data–models (DDM) paradigm: systems that support, rather than replace, human judgment by combining interactive dialogue, structured data, and analytical models. This human-in-the-loop conception has proven durable; contemporary reappraisals argue that its core logic remains valid even amid big data analytics and cognitive computing, suggesting that DSS theory anticipated—without fully specifying—the tensions that automated analytics would eventually introduce between system autonomy and human judgment. The critical continuity is that Sprague's framework presumed an active human dialogue partner; augmented analytics systems, by contrast, increasingly compress or eliminate that dialogue step, generating insights and narratives with minimal iterative human query formulation.

This shift was formally named in 2017, when Gartner analysts introduced the term "augmented analytics" to describe the use of machine learning and artificial intelligence to automate data preparation, insight generation, and insight explanation within analytics and BI platforms. Gartner characterized this as the third distinct wave of BI evolution and predicted, accurately in retrospect, that machine-driven insight generation would become the durable industry default rather than a transitional phase. Subsequent peer-reviewed treatments of augmented analytics have elaborated this definition: Alghamdi and Al-Baity (2022), publishing in Sensors, describe augmented analytics as integrating machine learning and natural language processing to automate analytical workflows for both technical and non-technical ("citizen data scientist") users, while cautioning that AI-driven analytics cannot fully substitute for human decision-making because most organizational problems resist purely automated resolution. This caveat is analytically important: it implicitly acknowledges that automated insight generation introduces a quality-control problem that vendors' efficiency narratives tend to elide.

2.2 Automated Insight Discovery and the Interestingness Problem

A parallel and more computationally oriented literature has developed within data mining and knowledge discovery research, concerned with how to identify which patterns, among the enormous number an algorithm can detect in any sufficiently large dataset, are worth surfacing to a human user. Geng and Hamilton's (2006) widely cited survey in ACM Computing Surveys remains the most comprehensive treatment of this problem, cataloguing objective measures (support, confidence, lift, conviction, peculiarity) and subjective measures (novelty relative to prior user belief, actionability for a specific decision) of "interestingness," and explicitly noting that good interestingness measures must simultaneously reduce the computational and cognitive costs of pattern discovery while ensuring that surfaced patterns are genuinely useful rather than merely statistically present.

Building on this foundation, Ding et al. (2019) proposed QuickInsights, a Microsoft Research system presented at the ACM SIGMOD conference, which operationalized automated insight mining over multidimensional tabular data by ranking candidate insights according to statistical impact and significance while using subspace pruning techniques to maintain computational tractability at scale. QuickInsights demonstrates that the industry's automated insight-generation engines operationalize interestingness primarily through statistical significance and effect size—criteria that map cleanly onto only one of Geng and Hamilton's (2006) proposed dimensions while leaving contextual relevance and organizational actionability comparatively under-operationalized. This is a critical gap: a pattern can be statistically striking while being wholly irrelevant to the decision a manager is actually trying to make, or can restate a fact already well understood by domain experts.

2.3 Insight Quality in Visual Analytics: Provenance, Perception, and User Judgment

A third relevant literature emerges from human-computer interaction and visual analytics research, which has approached "insight" not as an algorithmic output but as a cognitive event occurring within a human analyst. Gotz and Zhou (2009), publishing in Information Visualization, developed an influential framework for characterizing users' visual analytic activity to support "insight provenance"—a historical record of the reasoning process by which a human analyst arrives at a conclusion. Their work treats insight as fundamentally relational: an insight's value is partly constituted by the process and context through which a specific user derived it, not solely by properties intrinsic to the underlying data pattern. This stands in notable tension with the data-mining tradition, which treats insight quality as a largely data-intrinsic, computable property. More recent work has begun to bridge this divide empirically: Law, Endert, and Stasko (2020) investigated what professional visualization users actually consider to be a "data insight" in practice, while He et al. (2021), publishing in IEEE Transactions on Visualization and Computer Graphics, developed methods for characterizing insight quality through analysis of user interaction patterns during visual data exploration. Collectively, this literature suggests that insight quality is not a single computable scalar but a multidimensional and partly user-dependent construct.

2.4 Natural Language Generation and the Narration of Insights

Augmented analytics systems typically do not present raw statistical output; they narrate it. This narration layer draws on the natural language generation (NLG) literature, most comprehensively surveyed by Gatt and Krahmer (2018) in the Journal of Artificial Intelligence Research, which distinguishes "data-to-text" generation—producing natural language from non-linguistic structured input—as a distinct and long-standing NLG subfield. Gatt and Krahmer's survey documents a persistent and unresolved evaluation problem within NLG research itself: fluent, grammatically well-formed generated text is comparatively easy to produce and evaluate, but evaluating whether the generated text is faithful to the underlying data, non-misleading in its implied causal or comparative claims, and appropriately hedged given statistical uncertainty, remains a substantially harder and less standardized problem.

This is directly relevant to augmented analytics dashboards, where natural-language insight narration can convey unwarranted confidence, causal implication, or comparative significance that the underlying statistical pattern does not actually support. Relatedly, Rudin (2019), writing in Nature Machine Intelligence, cautions more broadly that explaining opaque model outputs after the fact is a poor substitute for models that are inherently interpretable—a concern with direct bearing on whether narrated automated insights can be trusted at face value.

2.5 Dashboard Effectiveness and BI Success: The Adoption-Centric Paradigm

A large and mature literature has examined BI dashboard design and effectiveness through the lens of information systems success and technology adoption. DeLone and McLean's (1992, 2003) IS Success Model—one of the most extensively applied theoretical frameworks in information systems research, encompassing system quality, information quality, service quality, use, user satisfaction, and net benefits—has been repeatedly applied and re-validated across BI contexts. Wixom and Watson (2001), in an empirical study of data warehousing success published in MIS Quarterly, found that system quality and data quality factors were significantly related to perceived net benefits, and that management support, resources, and user participation shaped implementation outcomes—an early empirical anchor for the broader IS success tradition applied specifically to BI infrastructure.

Krishnakumar's (2022) systematic review of BI tools and strategic dashboarding techniques similarly found that, despite widespread adoption of platforms such as Tableau, Power BI, and Looker, the literature offers only a "fragmented understanding" of their comparative effectiveness, and identified data quality issues, user adoption barriers, and the absence of data governance frameworks as recurring challenges. Applied case studies reinforce this pattern: Wibiayu and Siallagan (2022) documented the influence of BI dashboards on decision-making within a government agency context, while Ranjan and Foropon (2021), writing in the International Journal of Information Management, examined how big data analytics contributes to organizational competitive intelligence. Underlying much of this literature is the broader resource-based view of analytics capability articulated by Gupta and George (2016), who conceptualized big data analytics capability as an organizational resource combining tangible infrastructure, human skills, and intangible factors such as data-driven culture and organizational learning—a conceptualization this paper draws upon in defining organizational data maturity as a moderator (Section 7.2).

What is notable across this literature is its near-total focus on the dashboard as a presentation and interaction artifact—evaluated through constructs such as system quality, usability, and user satisfaction—rather than on the epistemic quality of the analytical content the dashboard displays. Even reviews that explicitly acknowledge "data quality issues" as a barrier to BI effectiveness are, on inspection, referring to input data quality rather than to the quality of any AI-generated analytical output layered atop that data. This is precisely the conflation this paper seeks to resolve.

2.6 Emerging Empirical Work on Trust, Cognition, and Augmented Analytics

Only very recently has empirical research begun to directly examine how augmented analytics capabilities affect the cognitive and behavioral experience of non-technical BI users. Pham Thi Phuong et al. (2026), in a working paper applying partial least squares structural equation modeling to a sample of 250 business professionals, found that augmented analytics capabilities significantly increased perceived ease of use, perceived usefulness, and trust in BI systems, and that trust exerted a direct, positive effect on self-reported decision quality. While this is a valuable early empirical contribution, it also illustrates a limitation common to this nascent stream: decision quality is measured through user perception and self-report rather than through any independent evaluation of whether the underlying automated insights were themselves statistically valid, contextually relevant, or non-trivial. This is not a flaw in the cited study's design so much as a structural blind spot in the broader research stream to which it contributes: perception-based measures of augmented analytics' value cannot, by construction, detect the failure mode this paper is centrally concerned with—confidently narrated, plausible-sounding, but epistemically weak automated insights.

2.7 Data Quality Versus Insight Quality: A Persistent Conflation

A final relevant thread concerns data quality research proper. Wang and Strong's (1996) foundational framework, developed from a data-consumer perspective in the Journal of Management Information Systems, established four now-canonical categories of data quality—intrinsic, contextual, representational, and accessibility—derived empirically from surveying what data consumers actually value beyond mere accuracy. This framework has proven durable and is still the most frequently cited data quality framework across subsequent big data and data governance research. Crucially, however, Wang and Strong's framework, like nearly all of the data quality literature it spawned, is concerned with the quality of input data—the raw material fed into an analytical process—not with the quality of analytical outputs subsequently derived from that data by an algorithm. A dataset can satisfy every dimension of the Wang and Strong framework and still yield automatically generated insights that are statistically spurious, contextually irrelevant, or practically actionless, because insight quality is a property of the analytical transformation applied to the data, not of the data itself. This distinction, while conceptually simple once stated, is not explicitly drawn anywhere in the reviewed literature, and its absence is a central symptom of the research gap this paper addresses. A related and complementary caution comes from the software-engineering literature on production machine learning systems: Sculley et al. (2015), in a widely cited NeurIPS paper on "hidden technical debt," documented how automated pipelines accumulate unseen risks—entangled dependencies, hidden feedback loops, and silent degradation—that are rarely visible from the output layer alone, reinforcing the case for an explicit, externally imposed insight-quality check rather than an assumption that a well-engineered pipeline is self-auditing.

2.8 Critical Synthesis

Table 1 synthesizes this review across the literature's four constituent domains, comparing their respective units of analysis, dominant methods, and—most importantly for this paper's purpose—their treatment (or non-treatment) of insight quality as a distinct evaluative construct. This synthesis makes the paper's central claim visible in tabular form: five of six reviewed literature domains either omit insight quality as a distinct construct entirely or operationalize it in a way that is incompatible with, or unconnected to, the other domains' treatment of the same underlying idea.

Table 1. Comparative Synthesis of Literature Domains Relevant to Insight Quality in Augmented Analytics

Literature Domain	Representative Sources	Unit of Analysis	Treatment of "Insight Quality"	Key Limitation for This Study
DSS / BI Foundations	Sprague (1980); DeLone & McLean (1992, 2003); Wixom & Watson (2001)	The information system as a whole	Implicit, subsumed under "information quality" construct	Predates automated insight generation; assumes active human dialogue
Data Mining / Interestingness	Geng & Hamilton (2006); Ding et al. (2019)	The algorithmically discovered pattern	Explicit but narrowly statistical (support, lift, significance)	Limited incorporation of organizational/contextual relevance
Visual Analytics / Insight Provenance	Gotz & Zhou (2009); Law et al. (2020); He et al. (2021)	The human analyst's cognitive process	Explicit but user- and process-dependent	Difficult to operationalize at scale of automated dashboards
NLG / Data Narration	Gatt & Krahmer (2018); Rudin (2019)	The generated text artifact	Implicit; concerned with faithfulness, not decision relevance	Faithful narration of a poor insight is still a poor insight
BI Dashboard Effectiveness	Krishnakumar (2022); Wibiayu & Siallagan (2022); Ranjan & Foropon (2021); Gupta & George (2016)	The dashboard as artifact/system	Absent; "data quality" refers to input data only	Conflates presentation quality with analytical content quality
Augmented Analytics & Trust	Alghamdi & Al-Baity (2022); Pham Thi Phuong et al. (2026)	The user's perception and behavior	Absent; decision quality measured via self-report	Cannot detect confidently-presented but objectively weak insights

3. RESEARCH GAP

The literature reviewed in Section 2 reveals a specific and consequential gap rather than a generic call for "more research." Three observations, taken together, define this gap precisely. First, the data mining and knowledge discovery literature has produced mature, computationally operational measures of pattern "interestingness" (Geng & Hamilton, 2006; Ding et al., 2019), but these measures are overwhelmingly statistical in character while incorporating organizational context, task relevance, and managerial actionability only weakly or implicitly. A pattern can score highly on every conventional interestingness metric while being of negligible value to the specific decision a manager is trying to make. Second, the visual analytics and human-computer interaction literature (Gotz & Zhou, 2009; Law et al., 2020; He et al., 2021) has established that insight, understood as a cognitive event, is partly constituted by the human analyst's reasoning process and context—a conception that is difficult to reconcile with, or translate into, the automated and largely human-independent insight-generation pipelines that augmented analytics platforms now deploy at scale. Third, the BI dashboard effectiveness literature (Krishnakumar, 2022; Wibiayu & Siallagan, 2022; Ranjan & Foropon, 2021) and the emerging augmented-analytics adoption literature (Alghamdi & Al-Baity, 2022; Pham Thi Phuong et al., 2026) evaluate BI systems predominantly through system quality, usability, adoption, and user-perceived trust or decision quality. None of these studies independently evaluates whether the automatically generated insights driving these perceptions are themselves statistically valid, non-trivial, or contextually relevant.

The consequence of these three gaps, taken jointly, is that no existing framework allows a researcher or practitioner to ask—let alone answer—the question: is this specific AI-generated insight, currently displayed on this dashboard, actually good? Existing data quality frameworks (Wang & Strong, 1996) evaluate the wrong layer of the analytics pipeline. Existing interestingness measures evaluate the right layer but on an incomplete set of criteria. Existing BI effectiveness research evaluates system-level and perceptual outcomes that are causally downstream of, and therefore no substitute for, insight quality itself. This paper's contribution is to close this specific integrative gap: the absence of a multidimensional, theoretically grounded framework for evaluating the quality of insights generated by AI-augmented BI systems, situated explicitly between the input-data-quality layer and the user-perception layer, rather than collapsed into either.

Table 2. Research Gap Matrix

Evaluative Dimension	Data Mining Literature	Visual Analytics Literature	BI/Dashboard Literature	Augmented Analytics/ Trust Literature
Statistical validity	Yes — extensively	Partially — implicit in provenance tracking	No	No
Contextual/task relevance	Weakly — proxied by significance	Yes — central to provenance concept	Partially — via usability/task-fit	Partially — via perceived usefulness
Novelty (vs. prior knowledge)	Yes — as "surprisingness"	Yes — as user-relative novelty	No	No
Actionability	Weakly — theorized, rarely measured	Partially — via decision outcomes	No — treated as system outcome	Yes — but only via self-report

As Table 2 illustrates, no single literature domain—nor any combination of the domains reviewed—currently provides integrated coverage across all four evaluative dimensions simultaneously. This fragmentation is the precise gap the proposed Insight Quality Evaluation (IQE) framework is designed to address.

4. RESEARCH QUESTIONS

Building directly on the research gap articulated in Section 3, this study addresses the following research questions:

- **RQ1:** What conceptual dimensions constitute "insight quality" when insights are automatically generated by AI-augmented business intelligence systems, as distinct from the quality of the underlying input data?
- **RQ2:** How do statistical validity, contextual relevance, novelty, and actionability interact as evaluative dimensions of automatically generated insights, and are these dimensions independent or do they trade off against one another?
- **RQ3:** How do moderating factors—specifically task complexity and organizational data maturity—shape the relationship between augmented analytics system capabilities and the quality of insights they generate?
- **RQ4:** What methodological approach can feasibly and rigorously evaluate insight quality along these dimensions, given that insight quality is partly computable (statistical validity, novelty) and partly context- and user-dependent (relevance, actionability)?

5. RESEARCH OBJECTIVES

Corresponding to the research questions above, this study pursues four objectives:

- **RO1:** To synthesize and critically integrate literature from data mining, visual analytics, business intelligence, and augmented analytics research in order to define insight quality as a distinct, multidimensional analytical construct, situated explicitly between input data quality and user-perceived decision quality.
- **RO2:** To develop an integrative conceptual framework—the Insight Quality Evaluation (IQE) framework—that specifies statistical validity, contextual relevance, novelty, and actionability as core evaluative dimensions of AI-generated insights within BI dashboard environments.
- **RO3:** To theorize task complexity and organizational data maturity as moderating factors that condition the relationship between augmented analytics capabilities and insight quality outcomes, drawing on established theoretical perspectives from decision support systems and information systems success research.
- **RO4:** To propose a rigorous, feasible mixed-methods research design—combining computational evaluation of insight-generation algorithms against public benchmark datasets with structured expert-panel judgment—that future researchers can operationalize to empirically test and refine the IQE framework.

6. THEORETICAL BACKGROUND

The Insight Quality Evaluation (IQE) framework proposed in this paper does not emerge from a single theoretical tradition but is deliberately assembled from four complementary theoretical perspectives, each addressing a different facet of the problem: bounded rationality theory, decision support systems theory, information systems success theory, and data mining interestingness theory. This section develops each in turn and shows how, together, they motivate the four-dimensional structure of the proposed framework.

6.1 Bounded Rationality and the Rationale for Analytical Delegation

Simon's (1947) theory of bounded rationality provides the foundational behavioral rationale for augmented analytics itself. Simon argued that human decision-makers, constrained by limited memory, attention, and computational capacity, cannot achieve the complete rationality assumed by classical decision theory; instead, they "satisfice"—selecting a course of action that is good enough relative to some threshold, rather than optimal relative to an exhaustive search of the decision space. Augmented analytics systems are, in this light, best understood as a technological response to bounded rationality—an attempt to extend the effective analytical capacity of the human decision-maker by delegating pattern search to a machine.

This framing carries an important implication frequently overlooked in vendor and practitioner discourse: delegation under bounded rationality does not eliminate the satisficing problem; it relocates it. Where a human analyst previously satisficed by choosing which patterns to investigate given limited time, an automated insight-generation algorithm now satisfices on the analyst's behalf, using whatever criteria are embedded in its ranking function. If those embedded criteria are narrow, then the algorithm's "satisficing" choices may systematically diverge from what a boundedly rational but domain-expert human would have selected as good enough.

6.2 Decision Support Systems Theory and the Dialogue-Data-Models Paradigm

Sprague's (1980) DSS framework offers the second theoretical anchor. Sprague characterized effective decision support around a dialogue–data–models (DDM) paradigm, in which a human decision-maker engages in iterative dialogue with a system that combines organized data with analytical models. This continuity is theoretically important, but so is a key discontinuity: Sprague's paradigm presumes an active, iterative human dialogue partner shaping the analytical process at each stage. Augmented analytics systems increasingly compress or bypass this dialogue: insights are pushed to the user pre-formed, often before any query has been posed at all. The DSS tradition therefore supplies this paper with a normative benchmark—effective decision support has historically been understood as preserving meaningful human interpretive engagement with data—against which the more autonomous, insight-push architecture of augmented analytics can be theoretically evaluated.

6.3 Information Systems Success Theory and the Position of Insight Quality

The DeLone and McLean (1992, 2003) IS Success Model supplies the third theoretical anchor, and helps position insight quality precisely within the causal chain the model proposes. In its updated form, the model links system quality and information quality to use and user satisfaction, which in turn produce net benefits at the individual and organizational level—a relationship empirically substantiated in the BI context by Wixom and Watson (2001). However, when a BI system autonomously generates novel analytical claims rather than merely presenting existing data, "information quality" as classically defined does not fully capture what is being evaluated: the informational content did not exist, in the form presented, until the system's algorithm produced it. This paper argues that insight quality should be understood as a distinct antecedent construct sitting causally upstream of, but analytically separable from, the information quality construct in the DeLone and McLean model.

6.4 Interestingness Theory and the Incompleteness of Statistical Operationalization

The fourth theoretical anchor comes directly from data mining theory. Geng and Hamilton's (2006) synthesis of interestingness measures identifies conciseness, peculiarity, surprisingness, and actionability as conceptually distinct components of what makes a discovered pattern valuable—explicitly distinguishing objective measures, computable directly from data, from subjective measures, which require some model of user knowledge or goals to evaluate. This theoretical distinction is, in effect, the direct ancestor of this paper's proposed framework: statistical validity and novelty correspond reasonably well to the objective side of Geng and Hamilton's typology, while contextual relevance and actionability correspond to the subjective side, requiring some representation of organizational context that a purely data-driven algorithm cannot supply on its own.

6.5 Narrative Faithfulness as a Cross-Cutting Theoretical Concern

A final, cross-cutting theoretical thread comes from the natural language generation literature. Gatt and Krahmer's (2018) survey identifies faithfulness—the degree to which generated natural language accurately represents the underlying data without overstating certainty, causality, or significance—as a persistent and only partially solved evaluation problem in data-to-text generation, a concern echoed by Rudin's (2019) broader caution against relying on post-hoc explanations of opaque systems. This concern cuts across all four theoretical anchors described above: a statistically valid, contextually relevant, novel, and actionable insight can still be undermined at the point of consumption if its natural-language narration overstates what the underlying statistics support.

6.6 Integrating the Four Anchors

Table 3 summarizes how each theoretical anchor maps onto the dimensions of the conceptual framework developed in the next section.

Table 3. Theoretical Anchors and Their Contribution to the IQE Framework

Theoretical Anchor	Core Source(s)	Primary Contribution to Framework
Bounded rationality	Simon (1947)	Explains the behavioral rationale for delegating pattern search to automated systems, and why the "good enough" standard embedded in algorithms matters
Decision support systems (DDM paradigm)	Sprague (1980)	Estates the normative benchmark of preserved human interpretive engagement
IS Success Model	DeLone & McLean (1992, 2003); Wixom & Watson (2001)	Positions insight quality as a distinct antecedent construct upstream of information quality
Interestingness theory	Geng & Hamilton (2006); Ding et al. (2019)	Supplies the objective/subjective distinction underlying the framework's four dimensions
NLG faithfulness	Gatt & Krahmer (2018); Rudin (2019)	Identifies narrative faithfulness as a cross-cutting presentation-layer concern

7. CONCEPTUAL FRAMEWORK

Drawing on the theoretical anchors developed in Section 6, this paper proposes the Insight Quality Evaluation (IQE) framework, which conceptualizes insight quality as a multidimensional construct produced by the interaction of an augmented analytics system's insight-generation capabilities with two organizational moderating conditions, and which in turn shapes downstream decision-relevant outcomes.

7.1 Core Constructs

Statistical Validity refers to the degree to which an automatically generated insight reflects a genuine, non-spurious pattern in the underlying data, as opposed to an artifact of multiple comparisons, small sample sizes, confounding, or algorithmic overfitting. This construct draws directly on the objective interestingness measures catalogued by Geng and Hamilton (2006) and operationalized computationally by systems such as QuickInsights (Ding et al., 2019).

Contextual Relevance refers to the degree to which an insight bears on a decision, goal, or question that is actually salient to the organizational user encountering it, as opposed to being statistically real but practically peripheral. This construct draws on the subjective side of interestingness theory and on the insight-provenance literature's emphasis on the user- and task-dependence of insight value (Gotz & Zhou, 2009; Law et al., 2020).

Novelty refers to the degree to which an insight adds information beyond what the user, or the organization's existing analytical record, already knows—distinguishing a genuinely new observation from a restatement of an already well-understood seasonal effect or common domain knowledge.

Actionability refers to the degree to which an insight, if accepted, implies a specific, feasible organizational response—as opposed to describing a pattern with no clear behavioral or decision-relevant implication. This construct connects directly to the decision-outcome orientation of DSS theory (Sprague, 1980).

These four constructs are proposed as conceptually distinct and only partially correlated with one another—an insight can, in principle, score high on one dimension while scoring low on another. This non-collapsibility is a deliberate design feature of the framework: it directly addresses the research gap identified in Section 3, in which existing systems and literatures each implicitly privilege only one or two of these dimensions.

7.2 Moderating Factors

Task Complexity is proposed to moderate the relationship between augmented analytics system capabilities and insight quality outcomes. In low-complexity, well-structured decision tasks, automated statistical detection is more likely to align with contextual relevance and actionability, because the decision space is narrow and well-defined. In high-complexity, ill-structured strategic tasks, the gap between statistically detectable patterns and genuinely relevant, actionable insight is likely to widen, because relevance and actionability in such tasks depend on tacit organizational knowledge that automated systems do not typically represent.

Organizational Data Maturity is proposed as a second moderator, capturing the extent to which an organization has established data governance practices, high input-data quality (Wang & Strong, 1996), and analytical literacy among its BI user base—consistent with the resource-based view (Gupta & George, 2016). Organizations with low data maturity are likely to see augmented analytics systems produce insights that fail on statistical validity simply because the input data quality is poor.

7.3 Propositions

- **P1:** Augmented analytics system capabilities are positively associated with the volume of insights generated but are not, by default, positively associated with the average quality of those insights across all four IQE dimensions.
- **P2:** Statistical validity, contextual relevance, novelty, and actionability are empirically distinguishable dimensions of insight quality, exhibiting only moderate positive correlation with one another.
- **P3:** Task complexity negatively moderates the relationship between automated statistical pattern detection and contextual relevance/actionability, such that automated insight quality on these two dimensions degrades as task complexity increases.
- **P4:** Organizational data maturity positively moderates the relationship between augmented analytics capabilities and statistical validity, such that the same insight-generation algorithm produces higher-validity insights in higher-data-maturity organizational contexts.
- **P5:** Narrative faithfulness of NLG-based insight presentation moderates the relationship between an insight's underlying (objective) quality and a decision-maker's perceived insight quality, such that unfaithful or overstated narration can cause perceived quality to diverge from actual, underlying quality in either direction.
- **P6:** Insight quality, as jointly determined by P1–P5, is positively associated with downstream decision quality, but this relationship is attenuated where decision-makers cannot distinguish high- from low-quality automated insights at the point of consumption.

7.4 Framework Diagram Description

Figure 1. The Insight Quality Evaluation (IQE) Framework. The path diagram outlines the components left to right: Augmented Analytics System Capabilities -> Central Cluster (Statistical Validity, Contextual Relevance, Novelty, Actionability) -> Decision Quality / Organizational Outcomes, with top and bottom moderators (Task Complexity, Organizational Data Maturity) and presentation layer (NLG Narrative Faithfulness).

7.5 Variable Definitions

Table 4. Variable Definitions for the IQE Framework

Variable	Type	Conceptual Definition	Illustrative Operationalization
Statistical Validity	Core construct (dependent)	Degree to which an insight reflects a genuine, non-spurious data pattern	Effect size; multiple-comparison–corrected significance; cross-sample robustness
Contextual Relevance	Core construct (dependent)	Degree to which an insight bears on a salient organizational decision or goal	Expert-panel rating against a defined decision scenario
Novelty	Core construct (dependent)	Degree to which an insight adds information beyond existing organizational knowledge	Comparison against a prior-knowledge baseline; expert novelty rating
Actionability	Core construct (dependent)	Degree to which an insight implies a feasible, specific organizational response	Expert rating of implied action clarity and feasibility
Augmented Analytics System Capabilities	Independent variable	Automated pattern-detection and NLG-narration functionality	Presence/sophistication level of automated insight features
Task Complexity	Moderator	Structuredness and decision-space breadth of the analytical task	Categorized (low/medium/high) via task-complexity typologies
Organizational Data Maturity	Moderator	Level of data governance, input data quality, and analytical literacy	Composite index (Wang & Strong dimensions + governance maturity)
NLG Narrative Faithfulness	Presentation moderator	Degree to which narration accurately represents underlying findings	Faithfulness rating vs. underlying statistical output
Decision Quality	Outcome variable	Perceived/observed quality of decisions informed by the insight	Composite of self-reported confidence + outcome tracking

8. METHODOLOGY

Because this paper is conceptual in nature and does not report primary empirical data collection, this section specifies a rigorous, feasible mixed-methods research design that future researchers can operationalize to test the propositions advanced in Section 7. The design combines a computational evaluation strand, in which automated insight-generation algorithms are applied to public benchmark datasets and scored against objective IQE dimensions, with an expert-panel validation strand, in which the subjective IQE dimensions are assessed by domain and BI professionals.

8.1 Research Design

The proposed design proceeds in three phases. Phase 1 (Computational Evaluation) applies automated insight-generation algorithms — either open-source implementations approximating commercial approaches (e.g., subspace-search insight mining following Ding et al., 2019) or, where feasible, outputs from commercial augmented analytics tools — to selected public datasets, scoring each generated insight on statistical validity and novelty. Phase 2 (Expert Panel Validation) has a structured panel of BI practitioners and domain experts (recommended minimum n = 15–20 per dataset domain) rate a stratified sample of the algorithmically generated insights on contextual relevance and actionability, using a defined decision scenario specific to each dataset domain; narrative faithfulness is also assessed by comparing each insight's natural-language narration against its underlying statistical output. Phase 3 (Moderator Testing) stratifies the dataset sample along the two proposed moderators — task complexity and organizational data maturity (the latter operationalized via controlled data degradation) — to test Propositions P3 and P4 under controlled conditions.

8.2 Recommended Datasets

The following datasets are recommended, selected for domain diversity and established use in prior BI/analytics research:

- **UCI Machine Learning Repository:** retail/e-commerce transaction datasets and healthcare operational datasets, suitable for routine/operational-task scenarios.
- **Kaggle (academic-use-licensed datasets):** sales and supply-chain datasets, suitable for both routine and strategic decision scenarios.
- **World Bank Open Data:** macroeconomic and development indicators, suitable for strategic, ill-structured decision scenarios.
- **WHO Open Data:** public health indicator datasets, suitable for high-stakes, high-complexity decision scenarios.
- **OECD Data:** cross-country economic and social indicators, suitable for comparative, strategic-level scenario construction.
- **Open Government Data portals:** operational public-sector datasets, suitable for government-agency BI scenarios (cf. Wibiayu & Siallagan, 2022).

8.3 Variables

The independent, dependent, and moderating variables are as specified in Table 4. The primary dependent variable in Phase 1 is a composite Statistical Validity Score and a Novelty Score. The primary dependent variables in Phase 2 are panel-rated Contextual Relevance and Actionability scores, and a Narrative Faithfulness score.

8.4 Data Preprocessing

Preprocessing follows standard data mining pipeline conventions: missing-value handling appropriate to each dataset; outlier documentation (retained, not removed, since outlier-driven patterns are relevant to the "peculiarity" dimension); standardization of variable types and units across datasets; and construction of a "known pattern baseline" for each dataset through structured literature and documentation review, required to operationalize the Novelty Score.

8.5 Statistical and Computational Methods

For Phase 1, subspace-search or association-rule-based insight mining algorithms (following Geng & Hamilton, 2006, and Ding et al., 2019) are applied, followed by multiple-comparison correction (e.g., Benjamini–Hochberg false discovery rate control) to guard against the spurious-pattern inflation risk inherent in exhaustive automated pattern search — a risk formally characterized in the "garden of forking paths" problem (Gelman & Loken, 2013). For Phase 2, inter-rater reliability is assessed via intraclass correlation coefficients, and construct distinctiveness across the four IQE dimensions (testing P2) is assessed via confirmatory factor analysis. For Phase 3, moderation effects (P3, P4) are tested via multi-group comparison or interaction-term regression models.

8.6 Validation Methods

Three validation mechanisms are proposed: inter-rater reliability for all expert-panel-derived scores (pre-registered threshold, e.g., ICC >= 0.70); cross-dataset replication, requiring core findings to replicate directionally across at least three of the recommended dataset domains; and construct validity testing via confirmatory factor analysis to empirically verify that the four proposed IQE dimensions load as distinct factors rather than collapsing into a single general factor.

8.7 Ethical Considerations

Because the proposed design relies exclusively on publicly available, non-personal, aggregate datasets and on expert panelists participating as professional consultants, the primary ethical considerations are: standard informed consent and voluntary participation protocols for expert panelists; transparency regarding any commercial augmented analytics tools used in Phase 1, including disclosure of vendor involvement to avoid conflicts of interest; and careful framing of decision scenarios used in Phase 2, particularly those drawn from health (WHO) data, to avoid presenting panelists with scenarios that could be mistaken for real clinical decision support.

8.4 Methodology Summary Table

Table 5. Methodology Summary

Phase	Objective	Data Source	Method	Primary Output
Phase 1: Computational Evaluation	Assess statistical validity and novelty	UCI, Kaggle, World Bank, WHO, OECD, Open Gov Data	Subspace-search/association-rule mining; multiple-comparison correction	Statistical Validity Score; Novelty Score
Phase 2: Expert Panel Validation	Assess relevance, actionability, faithfulness	Sampled Phase 1 insights + decision scenarios	Structured panel rating (n=15–20/domain); inter-rater reliability	Relevance, Actionability, Faithfulness Scores
Phase 3: Moderator Testing	Test task complexity and data maturity as moderators	Stratified scenario design + controlled degradation	Multi-group comparison; interaction regression; CFA	Support/non-support for P3, P4; validated factor structure

9. DATA ANALYSIS APPROACH

This section elaborates the analytical workflow through which the three-phase methodology described in Section 8 would be executed, moving from raw automated insight output through to a validated empirical test of the IQE framework's propositions.

9.1 Overview of the Analytical Workflow

The analytical workflow proceeds through four connected stages:

- **Stage 1 — Insight Generation and Statistical Scoring:** automated insight-mining algorithms are run against each recommended dataset, producing a candidate pool of insights, each scored computationally on Statistical Validity and Novelty before any human involvement.
- **Stage 2 — Stratified Sampling for Human Evaluation:** because the candidate pool is typically far too large for exhaustive expert review, a stratified sample is drawn across the Statistical Validity x Novelty score space, ensuring the expert panel evaluates insights spanning the full range of computationally scored quality rather than only the algorithm's top-ranked outputs.
- **Stage 3 — Expert Panel Scoring and Reliability Analysis:** the stratified sample is presented to the domain expert panel, blind to Stage 1 scores, who rate Contextual Relevance, Actionability, and Narrative Faithfulness.
- **Stage 4 — Integration and Hypothesis Testing:** computational and expert panel scores are merged at the level of the individual insight, producing the full four-dimensional IQE profile for each sampled insight, supporting confirmatory factor analysis (P2), moderation analysis (P3–P4), and mediation analysis (P5).

9.2 Quantitative Analysis Techniques

Construct distinctiveness (P2) is tested via confirmatory factor analysis, comparing model fit between a four-factor model and a single-factor model using standard fit indices and a chi-square difference test. Moderation by task complexity and data maturity (P3, P4) is tested via multi-group structural equation modeling or interaction-term moderated regression. Narrative faithfulness effects (P5) are tested via moderated mediation analysis. Insight volume versus quality (P1) is tested via correlational analysis comparing total insight volume against mean IQE profile. Each analysis is repeated independently across at least three of the six recommended dataset domains before any pooled synthesis is attempted.

9.3 Qualitative Analysis Component

In addition to quantitative panel ratings, the expert panel protocol should capture open-ended justifications for each relevance and actionability rating, analyzed thematically with dual-coder agreement checks, to surface recurring reasons why algorithmically well-scored insights were judged low-relevance or low-actionability by domain experts, and vice versa.

9.4 Supporting Technical Architecture

The architecture consists of standard pipeline designs mapping the components: Figure 2 (Data Pipeline Architecture from Ingestion to Preprocessing, Mining, Scoring, and Integration Layer), Figure 3 (Analytics Workflow separating computational and human steps), and Figure 4 (Dashboard Instrumentation Layer implementing real-time scoring and quality badges).

9.5 Software and Tools

Consistent with the methodological specification in Section 8, recommended tool categories include open-source statistical computing environments (R or Python, including packages for confirmatory factor analysis and multiple-comparison correction) for Stages 1 and 4; structured survey/rating platforms for Stage 3 expert panel data collection; and qualitative analysis software for the thematic coding described in Section 9.3.

10. RESULTS

As stated at the outset of this paper, this study is conceptual in nature: it proposes a theoretical framework and a research design for future empirical validation, rather than reporting findings from primary data collection. Consistent with the requirement to present empirical results only where genuinely supported by real data, no statistical findings, experimental outcomes, or dataset-derived figures are reported in this section, and none should be inferred from the illustrative material below.

To make the framework and methodology concrete, this section offers a purely illustrative walkthrough of how the IQE framework and the analytical workflow described in Section 9 would be applied to a hypothetical scenario. All figures, scores, and outcomes in this walkthrough are schematic placeholders included solely to demonstrate the mechanics of the proposed evaluation process; they do not represent actual computed values, real dataset outputs, or claimed empirical findings, and should not be cited as such.

10.1 Illustrative Scenario

Consider a hypothetical retail organization that has deployed an augmented analytics dashboard drawing on a UCI-repository-style transactional sales dataset. The system's automated insight-mining engine generates a candidate pool of insights — for example, detected anomalies in weekly sales by product category, correlations between promotional activity and basket size, and segment-level trend breaks. Each candidate insight is assigned a computational Statistical Validity score and a Novelty score.

10.2 Illustrative Application of the Stratified Sampling Step

A stratified sample would be drawn across the full range of computationally scored insights — not only the algorithm's top-ranked outputs. In this hypothetical illustration, the sample might include: an insight with high statistical validity and high novelty; an insight with high statistical validity but low novelty (e.g., a well-known seasonal effect); and an insight with only moderate statistical validity but flagged by the algorithm as high-impact due to a large, if noisy, effect size.

10.3 Illustrative Application of Expert Panel Scoring

A hypothetical expert panel would rate each sampled insight against a defined decision scenario on Contextual Relevance, Actionability, and Narrative Faithfulness. Based purely on the theoretical propositions developed in Section 7 (not on any actual data): the high-validity, high-novelty insight would plausibly receive high panel ratings on both relevance and actionability; the high-validity, low-novelty insight would plausibly receive lower novelty and relevance ratings, since experienced managers would likely already anticipate a well-known seasonal pattern; and the moderate-validity, high-computed-impact insight illustrates the risk that a purely statistical ranking algorithm might surface an insight prominently due to raw effect size that a careful validity check and expert review would jointly downgrade.

10.4 Illustrative Integration and Interpretation

The merged computational and panel scores for this hypothetical sample would populate the four-dimensional IQE profile described in Section 7.1 for each insight, enabling the confirmatory factor analysis, moderation testing, and mediation analysis specified in Section 9.2. In an actual empirical application, this integrated profile would be the object of statistical hypothesis testing against Propositions P1–P6; here it serves only to show how the abstract constructs translate into a concrete, insight-by-insight evaluation record.

10.5 What This Illustration Does Not Show

This illustrative walkthrough does not, and cannot, demonstrate: whether the IQE framework's four dimensions actually behave as empirically distinct constructs in real data; whether task complexity or organizational data maturity actually moderate insight quality as proposed; or whether improved insight quality, as measured by the framework, actually produces better organizational decisions. These remain open empirical questions for the future research program outlined in Section 15.

11. DISCUSSION

This section interprets the conceptual and methodological contributions developed in Sections 6 through 10 against the research questions posed in Section 4, situates the IQE framework within ongoing debates in the augmented analytics literature, and addresses several tensions the framework surfaces but does not fully resolve.

11.1 Revisiting the Research Questions

RQ1 asked what conceptual dimensions constitute insight quality when insights are automatically generated. The theoretical synthesis in Section 6 and the framework in Section 7 answer this by proposing four dimensions — statistical validity, contextual relevance, novelty, and actionability — derived from the explicit, if previously unintegrated, treatment each dimension already receives in a distinct literature. The contribution here is integrative rather than inventive.

RQ2 asked how these four dimensions interact. The framework's Proposition P2 treats this as an open empirical question rather than assuming either full independence or full collapse into a single "quality" factor, and Section 10's illustrative walkthrough demonstrates concretely why this matters: a statistically valid, high-impact insight may still be low-quality overall once contextual and validity concerns are jointly weighed.

RQ3 asked how task complexity and organizational data maturity moderate the relationship between system capabilities and insight quality. The theoretical rationale developed in Sections 6.1 and 6.2 suggests these moderators operate through different mechanisms: task complexity is theorized to affect primarily the relevance and actionability dimensions, while organizational data maturity is theorized to affect primarily the statistical validity dimension, operating upstream at the level of input data quality. This distinction between where in the pipeline each moderator is theorized to bite is, to this paper's knowledge, not made explicit anywhere in the reviewed literature.

RQ4 asked what methodology could feasibly evaluate insight quality along dimensions that are partly computable and partly context-dependent. The three-phase mixed-methods design in Section 8, and its elaboration into a concrete analytical workflow in Section 9, demonstrates that this evaluation is achievable without requiring either reducing the entire construct to computable statistical properties alone, or relying purely on subjective user perception, which the review of Pham Thi Phuong et al. (2026) showed cannot distinguish objectively strong from objectively weak insights.

11.2 Tensions the Framework Surfaces

Tension 1: Automatability versus completeness. Two of the framework's four dimensions are reasonably automatable at scale, while the other two currently require human expert judgment. This paper does not claim to resolve this tradeoff; rather, it argues that making the tradeoff explicit and visible — for instance, through the dashboard "quality badge" architecture proposed in Section 9.4 — is preferable to the current default, in which the tradeoff exists but is invisible to the end user.

Tension 2: Whose relevance and whose actionability? The contextual relevance and actionability dimensions are defined relative to a decision-maker's goals, but real organizations contain multiple decision-makers with potentially conflicting goals. Future refinements of the framework would need to specify whether insight quality scores should be role-specific or role-aggregated within a single organizational dashboard.

Tension 3: The risk of false precision. A framework that formalizes insight quality into four scored dimensions carries an inherent risk of conferring unwarranted scientific authority on what remains, for two of its four dimensions, an expert judgment process with the same subjectivity limitations as any other expert-elicitation method. This risk is best managed through transparent reporting of measurement uncertainty rather than through avoiding formalization altogether.

11.3 Relating the Framework Back to Conflicting Findings in the Literature

The literature review identified an apparent tension between the perception-based findings of Pham Thi Phuong et al. (2026) and this paper's central concern that automated insights may often fail on objective quality grounds. These findings are not necessarily contradictory: it is entirely possible for augmented analytics capabilities to increase perceived usefulness and self-reported trust, while the insights driving those perceptions vary widely in objective statistical validity, relevance, and actionability. A productive direction for future research would explicitly test whether user trust tracks actual insight quality (well-calibrated trust) or is instead driven primarily by presentation fluency and narrative confidence regardless of underlying quality (miscalibrated trust).

11.4 Generalizability Considerations

The framework is proposed as domain-general, and the recommended dataset portfolio spans retail, healthcare, macroeconomic, and public-sector domains specifically to support testing this generality. However, organizations or sectors characterized by highly structured, rule-governed decision processes are theoretically expected to show smaller gaps between computationally scored and panel-scored insight quality than sectors characterized by ill-structured, judgment-intensive decisions — itself a testable implication of Proposition P3 rather than an assumption external to the framework.

12. PRACTICAL IMPLICATIONS

12.1 Implications for BI Platform Vendors

Vendors of augmented analytics platforms currently compete substantially on the volume and fluency of automatically generated insights and natural-language narration. This paper's central practical recommendation is that vendors should treat insight quality as a distinct, measurable, and marketable product attribute in its own right, rather than an implicit byproduct of insight volume or narrative polish. Concretely, platforms should expose the statistical basis of each automated insight to users who wish to inspect it; incorporate mechanisms for capturing organizational context so that novelty scoring can be calibrated against actual organizational knowledge; and evaluate natural-language narration against faithfulness criteria adapted from the NLG evaluation literature (Gatt & Krahmer, 2018), since fluent narration can mask, rather than reveal, weak underlying statistical support.

12.2 Implications for Organizational Decision-Makers and BI Governance

For organizations deploying augmented analytics dashboards, this paper's framework suggests a concrete governance practice: before automated insights are permitted to inform consequential decisions, organizations should establish a lightweight internal review analogous to the expert-panel phase described in Section 8.1, particularly for insights that inform high-stakes or strategic decisions where Proposition P3 predicts the largest gap between algorithmic ranking and genuine relevance. Organizational data maturity, per the Wang and Strong (1996) and Gupta and George (2016) traditions underlying this construct, is a well-established governance target already pursued by most mature BI programs; this paper's contribution is to make explicit that data maturity investments pay a second dividend beyond input data reliability — they also plausibly improve the statistical validity of any AI-generated insights layered atop that data.

12.3 Implications for BI/Analytics Practitioners and Citizen Data Scientists

For non-technical BI users, this paper's framework suggests a practical heuristic: users encountering an automated insight should ask, in sequence, (a) does this claim hold up if I imagine testing it against a different time period or subgroup; (b) did I already know this; (c) does this bear on a decision I am actually facing right now; and (d) if I accepted this, what would I concretely do differently. This four-question heuristic, directly derived from the framework's constructs, offers organizations a low-cost training intervention that does not require deploying the full expert-panel methodology.

12.4 Implications for Regulatory and Professional Standards Bodies

As augmented analytics becomes further embedded in regulated sectors, professional and regulatory bodies overseeing data-driven decision-making may find the IQE framework a useful starting point for developing sector-specific insight-quality audit standards, analogous to existing data quality audit standards built on the Wang and Strong (1996) tradition. The four-dimensional structure itself — rather than any single "trust score" — is arguably a more defensible basis for such standards than the single-dimension statistical significance criteria that automated insight-mining systems currently emphasize by default.

13. THEORETICAL CONTRIBUTIONS

13.1 Establishing Insight Quality as a Distinct Construct

The most fundamental theoretical contribution of this paper is definitional: it establishes insight quality, in AI-augmented BI contexts, as a construct conceptually and causally distinct from both input data quality (Wang & Strong, 1996) and user-perceived decision quality (Pham Thi Phuong et al., 2026). By positioning insight quality as a distinct antecedent construct within the causal structure of the DeLone and McLean (1992, 2003) IS Success Model, this paper offers IS researchers a more precise vocabulary for locating exactly where, in an increasingly automated analytics pipeline, quality can be gained or lost.

13.2 Bridging Previously Disconnected Literatures

This paper's second theoretical contribution is to make explicit the connections between data mining interestingness theory, visual analytics insight-provenance research, NLG faithfulness research, and BI adoption research, which have developed largely independently despite bearing directly on the same underlying applied phenomenon. No prior framework, to this paper's knowledge, has drawn these specific theoretical threads together into a single applied evaluative structure.

13.3 Extending Bounded Rationality Theory to Automated Analytical Delegation

Section 6.1 extended Simon's (1947) bounded rationality theory to explain a second-order phenomenon: the satisficing behavior embedded in automated systems acting as analytical delegates on a bounded-rational human's behalf. This reframes the augmented analytics quality problem as a specific, contemporary instance of a long-standing organizational behavior problem — the delegation of judgment under resource constraints.

13.4 Revising the Sprague DDM Paradigm for Insight-Push Architectures

This paper's fourth theoretical contribution proposes that DSS theory's normative core — that effective decision support preserves meaningful human interpretive engagement — can be retained even under insight-push architectures, provided that engagement is relocated rather than eliminated: from the iterative query dialogue Sprague (1980) originally theorized to the point of insight consumption, through mechanisms such as the quality-badge dashboard architecture proposed in Section 9.4 and the four-question user heuristic proposed in Section 12.3.

14. LIMITATIONS

This is a conceptual framework and research-design proposal, not an empirical study. No data have been collected, no algorithms have been run against the recommended datasets, and no propositions (P1–P6) have been tested. The illustrative walkthrough in Section 10 should not be mistaken for, or cited as, empirical evidence.

Contextual relevance and actionability are, by design, not fully automatable and depend on structured expert judgment. Any organization or research team seeking to operationalize the full IQE framework faces the cost and logistical complexity of recruiting and managing qualified expert panels.

The framework's moderator propositions (P3, P4) themselves predict that findings may not generalize uniformly across sectors — meaning that even a fully executed empirical validation using the recommended datasets would likely need substantial further replication before the framework could be considered validated for sectors not represented in the initial validation studies.

This paper has deliberately simplified faithfulness into a single moderating construct for tractability; the NLG evaluation literature (Gatt & Krahmer, 2018) documents faithfulness evaluation as a multidimensional problem encompassing factual consistency, appropriate hedging, and avoidance of unwarranted causal implication.

The proposed methodology and analytical workflow are cross-sectional in design and do not address how insight quality, or the moderating effects of task complexity and data maturity, might evolve as an organization's augmented analytics system is used over time.

Direct algorithmic access to proprietary commercial augmented analytics engines is not generally available to academic researchers. Any empirical validation using open-source approximations may not fully generalize to the proprietary commercial systems most widely deployed in organizational settings — a limitation compounded by the kind of hidden, hard-to-audit technical debt documented in production machine learning systems more broadly (Sculley et al., 2015).

This paper's framework is focused specifically on epistemic insight quality and does not address a related but analytically distinct set of concerns regarding algorithmic bias, fairness, or disparate impact in automated insight generation (cf. Barocas & Selbst, 2016) — for instance, whether automated insight-mining algorithms are more likely to surface patterns affecting some organizational subgroups than others in ways that raise equity concerns.

This is a deliberate scope decision, flagged explicitly here as an acknowledged boundary of this paper's contribution.

15. FUTURE RESEARCH

The most immediate priority is execution of the three-phase mixed-methods design specified in Section 8. An initial validation study focused on a single dataset domain would represent a feasible first step, with cross-domain replication as a subsequent phase.

Future research should investigate whether faithfulness sub-dimensions — factual consistency, uncertainty hedging, and causal-implication avoidance — function as distinct moderators of the relationship between underlying and perceived insight quality (P5), potentially requiring their own dedicated evaluation protocol.

Future work should examine how sustained exposure to varying insight quality affects user trust calibration over time, testing whether users' capacity to distinguish high- from low-quality automated insights improves, deteriorates, or remains static with prolonged system use.

A natural and high-value future research direction is an integrative study directly testing whether user trust in automated insights tracks the IQE framework's objective quality dimensions (well-calibrated trust) or is instead driven primarily by narrative fluency and presentation confidence independent of underlying quality (miscalibrated trust).

Future research should examine whether automated insight-mining algorithms exhibit systematic differences in the volume or quality of insights surfaced across organizational subgroups or customer segments, integrating this paper's epistemic quality framework with the algorithmic fairness literature (Barocas & Selbst, 2016).

An ambitious but valuable future research direction is investigating whether large language models or other AI systems, provided with structured representations of organizational context, could provide a scalable proxy for expert-panel relevance and actionability judgments — validated against genuine human expert panels rather than assumed reliable.

16. CONCLUSION

The rapid embedding of artificial intelligence into business intelligence dashboards has fundamentally changed what a "dashboard" is: no longer merely a presentation surface for human-interpreted data, contemporary augmented analytics platforms actively generate, rank, and narrate analytical claims with minimal human involvement. This shift has proceeded largely unaccompanied by a corresponding evaluative framework capable of answering a deceptively simple question: is a given automatically generated insight actually good?

This paper has argued that existing literatures, examined individually, cannot answer this question. Data mining interestingness theory (Geng & Hamilton, 2006) supplies rigorous statistical operationalizations but only weakly incorporates organizational context. Visual analytics research on insight provenance (Gotz & Zhou, 2009; Law et al., 2020) rightly insists that insight is partly constituted by human interpretive process, but has not been translated into an evaluative standard applicable to largely autonomous insight-generation pipelines. BI dashboard effectiveness research (Krishnakumar, 2022; Ranjan & Foropon, 2021) and emerging augmented analytics trust research (Alghamdi & Al-Baity, 2022; Pham Thi Phuong et al., 2026) evaluate system-level adoption and user perception, but do not independently assess whether the insights driving those perceptions are themselves epistemically sound. And the mature data quality tradition established by Wang and Strong (1996) evaluates the wrong layer of the pipeline entirely.

In response, this paper has proposed the Insight Quality Evaluation (IQE) framework, theoretically grounded in bounded rationality theory (Simon, 1947), decision support systems theory (Sprague, 1980), information systems success theory (DeLone & McLean, 1992, 2003), and data mining interestingness theory (Geng & Hamilton, 2006), and specifying four evaluative dimensions — statistical validity, contextual relevance, novelty, and actionability — moderated by task complexity and organizational data maturity, and further shaped by narrative faithfulness at the point of presentation. The paper has further proposed a feasible, rigorous mixed-methods research design capable of empirically testing the six propositions the framework advances.

The framework's core theoretical claim — and its central practical implication — is that augmented analytics capabilities do not translate automatically or uniformly into decision quality. That translation is mediated by a multidimensional insight quality construct that current commercial systems and academic literatures alike have left substantially unmeasured. As organizations continue to scale their reliance on AI-generated analytical narratives across operational, financial, and strategic decision-making, the capacity to distinguish genuinely valid, relevant, novel, and actionable insights from statistically present but practically hollow ones is not a peripheral concern but a foundational precondition for augmented analytics to deliver on its stated promise. This paper offers a first, deliberately integrative step toward making that distinction measurable, testable, and actionable for researchers and practitioners alike.

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